Lab 2: Requirements & Specifications (Individual Portion)

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Table of Contents

List of Figures

No figures right now

List of Tables

No tables right now

1. Introduction

This document is the individual portion of the Software Requirements Specification (SRS) for Knowledge Is Power (KIP), a digital legal literacy platform for young adults. It is organized according to IEEE 830-1998 and covers the Introduction and the Overall Description.

1.1 Purpose

The purpose of this SRS is to define, in a structured and verifiable form, what the KIP system is required to do. It follows the IEEE 830-1998 recommended practice for software requirements specifications. The intended readers are the KIP development team, who will use the SRS as the basis for design and implementation, and the course instructor and other reviewers, who will evaluate whether the requirements are complete, consistent, and clear.

1.2 Scope

The software product described by this SRS is Knowledge Is Power (KIP), a web-based legal literacy platform for young people ages 16–19 (Macaluso, 2026). These users increasingly encounter laws involving driving, digital conduct, work, school, ownership, contracts, and safety without fully understanding the consequences of their choices, and law-related education remains largely absent from K-12 curricula (Centers for Disease Control and Prevention, 2026; Legal Services Corporation, 2017; National Highway Traffic Safety Administration, 2024; Touro Law Jacob D. Fuchsberg Law Center, 2017).

KIP will address this gap through short misconception-based quizzes and longer branching scenarios that ask users to apply laws to realistic situations. It will track each user's progress, recommend relevant modules and games based on performance, and connect lessons to official legal sources. The goals of KIP are to help young adults recognize legally significant situations, understand the potential consequences of their choices, and make more informed

decisions. KIP also serves as a supplemental educational aid for parents, teachers, and school administrators.

1.3 Definitions, Acronyms, and Abbreviations

CDC: Centers for Disease Control and Prevention

IEEE: Institute of Electrical and Electronics Engineers

JDBC: Java Database Connectivity

KIP: Knowledge Is Power

MFCD: Major Functional Component Diagram

SRS: Software Requirements Specification

UX: User Experience

1.4 References

Centers for Disease Control and Prevention. (2026). Transportation safety. Retrieved from https://www.cdc.gov/transportation-safety/index.html

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National Highway Traffic Safety Administration. (2024). Teen driving: Talk to your teen driver about safe driving. Retrieved from https://www.nhtsa.gov/teen-driving/parents-talk-yourteen-driver-about-safe-driving

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1.5 Overview

The remainder of this document describes KIP at a high level. Section 2, Overall Description, presents the product perspective (2.1), the main product functions (2.2), the characteristics of KIP's users (2.3), the constraints on the product (2.4), and the assumptions and dependencies that the requirements rely on (2.5). The specific requirements for each function are presented in the later sections of the full SRS.

2. Overall Description

This section describes the general factors that affect KIP and its requirements. It provides background for the specific requirements rather than stating them.

2.1 Product Perspective

KIP is a self-contained, web-based educational service built around learning modules, interactive games, user progress data, and source-backed legal content. Users access it through a modern web browser on a consumer device, and no additional software is required on the client.

The system is organized into four layers. The presentation layer, implemented with JavaScript, Bootstrap, and HTML5, provides the account portal, module browser, game browser, and scenario player. The application layer, implemented in Java with JDBC, provides authentication, the scenario engine, the feedback system, the curation system, and the source reference system. The data layer, implemented with SQLite, stores user, progress, module, game, and legal content data in five databases. The hardware layer consists of a virtual private server

that hosts the Java application and the SQLite database file, and the client devices used to reach it.

2.2 Product Functions

The main functions of KIP are summarized below.

Account access. Registered users can log in and create accounts. Newcomers can also play games as guests, with limited features, to try KIP before registering.

Module browsing. Users can browse legal learning modules such as traffic law, online conduct, school and work, and personal safety, and view their progress status for each topic.

"Fact or Fiction?" game. Users answer short questions about common legal beliefs and misconceptions with "true," "false," or "context-dependent."

"Choose the Outcome" game. Users explore a branching scenario by interacting with people, objects, and documents, make a decision, and see the consequences of that choice.

Feedback. After each decision, KIP indicates whether the answer was correct, incorrect, or context-dependent and explains the legal reasoning behind the result.

Progress tracking and recommendations. KIP records each user's topical strengths, weaknesses, and completion, and uses them to recommend games and modules. Users can still browse and select any available content.

Source references. Game content and feedback link to official legal sources, educational resources, and relevant vocabulary. KIP also tracks changes to legislation and flags content that may be outdated so administrators can review, revise, or disable it.

2.3 User Characteristics

KIP supports the following user roles.

Registered learner. The primary user is a young person ages 16–19 preparing to enter adulthood and navigate high-risk legal areas such as driving, work, and substance use. Learners are not expected to have any legal background, and they are comfortable using web-based technology. Registered learners have full access to games, modules, progress tracking, and recommendations.

Guest learner. A guest has the same background as a registered learner but plays without an account. Guests can try the games with limited features, and their progress is not retained.

Parent, teacher, or school administrator. These secondary stakeholders use KIP as a supplemental tool to reinforce legal awareness outside the classroom. They are not expected to have legal expertise.

Content administrator. Administrators review content that has been flagged as potentially outdated and can revise it or disable access to it.

2.4 Constraints

KIP must be delivered as a web-based service that works in a modern web browser, including on mobile devices, and requires an internet connection. The presentation layer must use JavaScript, Bootstrap, and HTML5; the application layer must use Java and JDBC; and persistent data must be stored in SQLite on a virtual private server. KIP's content must be tied to official legal sources so that users can verify where it comes from.

2.5 Assumptions and Dependencies

N/A